SUMAN LAMSAL

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Summary

Computer Science M.S. candidate at Indiana University with published research in deep learning for healthcare and hands-on work spanning LLM interpretability, causal inference, and computer vision. Built machine-unlearning pipelines to test reasoning faithfulness in domain-adapted LLMs, authored a fairness-constrained causal inference method, and delivered an HPC-trained pose estimation system now used by a neuroscience research group. Proficient in PyTorch, Hugging Face Transformers, and distributed training on Slurm-managed clusters. Seeking full-time research and ML engineering roles building reliable, interpretable AI systems.

Education

Master of Science in Computer Science — **Aug 2024 – Dec 2026**
Indiana University Indianapolis — *Indianapolis, IN | GPA: 3.78 / 4.0*

Bachelor of Engineering in Electronics, Communication & Information Engineering — **Nov 2019 – May 2023**
Tribhuvan University, Institute of Engineering (Pulchowk Campus) — *Lalitpur, Nepal | GPA: 3.88 / 4.0*

Publications

N. Bastakoti, S. Poudel, **S. Lamsal**, Y. Sharma, A. Jaiswal. "Heart Rate Estimation from PPG and Accelerometer via CNN-Transfo

- Uncertainty-aware architecture predicting heart rate mean and standard deviation under NLL loss; **MAE 4.82 bpm** (IEEE SPC 2015) and **6.20 bpm** (PPG-DaLiA) under Leave-One-Session-Out cross-validation at only **151K parameters 56× smaller** than Deep PPG (8.5M) with better accuracy.

Technical Skills

Languages: Python, SQL, R, Java, C/C++, Bash

ML / AI: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, DeepLabCut, OpenCV, LangChain

Research Methods: Machine unlearning (NPO), causal inference, pose estimation, model interpretability, LLM evaluation, transfer learning, statistical hypothesis testing

Infrastructure: Slurm / HPC clusters, Docker, Git, GitHub Actions, MLflow, Airflow, Linux, AWS (S3, EC2)

Data & Analysis: NumPy, Pandas, SciPy, Matplotlib, Seaborn, PostgreSQL, MySQL, Tableau, Jupyter

Research Experience

NLP Research: LLM Interpretability & Machine Unlearning — **Jan 2026 – May 2026**
Indiana University Indianapolis — *Indianapolis, IN*

- Built the **machine unlearning pipeline** (Negative Preference Optimization with KL regularization) used to test whether LLM chain-of-thought reasoning is causally linked to predictions implementing targeted MLP-layer selection, model checkpointing, and integration with the evaluation framework.
- Extended parametric faithfulness measurement (Tutek et al., EMNLP 2025) to domain-adapted models, evaluating **BioMistral-7B**, **Saul-7B**, and **Mistral-7B-Instruct** across **412 reasoning steps** on MedQA and LegalBench.
- Co-discovered **reasoning entrenchment**: domain-adapted reasoning is equally faithful (FF-HARD ∼67%, p=0.96 after controlling for unlearning efficacy) but resists parametric erasure in **70% of cases vs. 21%** for the base model, a distinction prior work conflated.
- Identified catastrophic forgetting as a domain-specific NPO failure mode: hyperparameters valid on medical text (Specificity 61%) collapsed the model on legal text (Specificity 16%), invalidating the run and demonstrating that unlearning requires domain-specific tuning.
- Ran distributed experiments on Indiana University's **Big Red 200** (NVIDIA A100-80GB); validated findings with bootstrap 95% confidence intervals, two-sample t-tests, and Cohen's d.

AI Research: Computer Vision for Biomedical Gait Analysis — **Jan 2026 – May 2026**
Indiana University Indianapolis (FADS Program) — *Indianapolis, IN*

- Sole-authored an end-to-end automated gait analysis pipeline for the project "Quantification of Gait in Down Syndrome Mouse Models using AI," collaborating with an interdisciplinary team of neuroscientists and computer scientists.
- Trained a **ResNet-50** pose estimation model (**DeepLabCut**, transfer learning) to **100,000 iterations** on 194 labeled frames, tracking six anatomical landmarks; final training loss **0.0023**, test error **8.75 px**.
- Automated model training on Indiana University's **Quartz HPC cluster** (NVIDIA V100, CUDA 11.8, **Slurm**), building a reusable submission tool handling project transfer, job-script generation, and Windows-to-Linux path conversion reducing training wall-clock time by over **95%** versus local execution.
- Engineered a quality-control layer (likelihood filtering, linear interpolation, temporal smoothing) converting noisy pose estimates into stable trajectories, plus an interactive pixel-to-centimeter calibration utility and a batch gait-parameter extractor.
- Validated all six extracted metrics against published biological reference ranges — velocity **11.34 cm/s**, hindpaw stride **3.04 cm**, bilateral asymmetry **1.0%** reducing per-video analysis from hours to minutes for the research group.

Professional Experience

Backend Engineer — **Aug 2023 – Aug 2024**
OCM Engineering Groups, LLC — *Greenwood, IN*

- Architected end-to-end data ingestion and validation pipelines for utility infrastructure inspection platforms, reducing manual data preparation time by **20%** across 3 client deployments.
- Developed automated GIS-based record-update processes and asset-management tooling, streamlining pole-inventory workflows and reducing data-entry error rates for utility and telecom clients.
- Maintained and extended backend APIs for data retrieval and reporting, ensuring reliability and scalability of mission-critical inspection data systems.

AI / ML Fellow — **Jan 2023 – Jun 2023**
Fusemachines — *Kathmandu, Nepal*

- Trained and evaluated ML models (Logistic Regression, Random Forest, CNNs) for predictive analytics tasks, improving classification accuracy by **15%** through systematic hyperparameter tuning and feature engineering.
- Built NLP and computer vision prototypes using **PyTorch** and **TensorFlow**, delivering working demos for text classification and image recognition use cases.
- Designed reproducible experiment pipelines with **Scikit-learn, Pandas, and MLflow**, standardizing preprocessing, feature selection, and evaluation so results could be independently reproduced.

Data Engineering Intern — **Jun 2022 – Nov 2022**
Everestwalks Groups — *Kathmandu, Nepal*

- Engineered and optimized ETL pipelines using **Python, SQL, and Pandas**, reducing data latency by **25%** and enabling near-real-time analytics availability.
- Automated data cleaning and transformation workflows with **Bash scripting** and **Apache Airflow**, eliminating manual intervention across 4 recurring data jobs.

Selected Projects

FairCFRNet: Causal Path-Specific Regularization for Fair Treatment Effect Estimation | *PyTorch, Causal Inference*

- Sole-authored a method integrating **differentiable path-specific fairness constraints** into CFRNet, addressing a gap in deep treatment effect estimation: existing methods balance distributions but cannot block specific unfair causal mechanisms.
- Designed a **Differentiable Causal Estimation** module computing path-specific effects inside the network's computational graph via AIPW estimation and counterfactual mediators, preserving end-to-end trainability.
- On the IHDP benchmark, cut the demographic parity gap **22%** (0.695 vs. 0.891) while *improving* PEHE **6.1%** (0.844 vs. 0.899) and raising ITE correlation from 0.367 to **0.445** demonstrating fairness-accuracy synergy rather than tradeoff.
- Characterized the full Pareto frontier across regularization strengths, including model collapse at λ=2.0 (correlation -0.088), establishing $\lambda \in [0.3, 0.7]$ as the usable deployment range.

Transformer vs. Mamba Architecture Comparison | *NLP, PyTorch, Hugging Face, Deep Learning*

- Benchmarked **BERT-base (110M)** against **Mamba-370M** on AI-generated vs. human-written text detection using a balanced **25K-sample** subset of a 347K-sample corpus; both reached **F1 = 0.99** (BERT 0.9954, Mamba 0.9922).
- Demonstrated Mamba's **linear-time scaling** advantage empirically: inference held flat at ~0.06s from 500 to 5,000 words while BERT climbed to ~1.1s roughly **18× faster** at long inputs despite 3.4× more parameters.
- Implemented end-to-end fine-tuning pipelines with custom classification heads, class-balanced sampling preserving the original length distribution, and evaluation via confusion matrices, F1, and AUROC on an **A100 GPU**.